LABSTREAM: SYSTEM ANALYSIS & ENHANCEMENTS

Diagnostic Laboratory & Test Result Portal | ITP 108 Event-Driven Programming

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1. EXECUTIVE SUMMARY & SCOPE REALIGNMENT

Labstream is an event-driven, web-based diagnostic laboratory management system designed for a single clinical laboratory. By incorporating the official project proposal approved by Prof. Jayson P. Joble (ITP 108 - Event-Driven Programming), the system architecture was realigned to prioritize academic objectives and eliminate artificial enterprise gaps.

- **Single-Laboratory Scope:** Explicitly aligned with Proposal Limitation #1. Complex multi-branch administrative modules were de-scoped in favor of robust single-clinic operations.
- **Supported Result Modalities:** Aligned with Proposal Limitation #4 (Text and numeric results with reference ranges). Complex antimicrobial susceptibility matrices and cytology narrative templates were deprioritized, focusing on scalar and panel laboratory testing.
- **Walk-In Cash Billing:** Aligned with Proposal Limitation #5 & #6 (In-person cashier settlement). Third-party payment gateways and HMO claim workflows were excluded in accordance with the proposal.
- **Manual Encoding Model:** Aligned with Proposal Limitation #3. Technologists encode values manually; hardware analyzer/LIS interfaces are omitted.

2. DEPLOYED ENVIRONMENT ROOT CAUSE ANALYSIS (RENDER)

A discrepancy was observed where the local development instance correctly retrieved and displayed staff accounts and catalog records, whereas the deployed Render production instance returned empty tables despite reporting a successful database connection.

- **Root Cause Diagnosis:** Render hosts the frontend as a static site and the backend as a distinct web service on different subdomains. The session cookie was configured with SameSite=Lax, preventing browsers from transmitting the HttpOnly authentication cookie on cross-origin fetch requests (credentials: include).
- **Secondary Vulnerability:** The requireSameOrigin middleware strictly inspects Origin headers. If FRONTEND_ORIGIN does not strictly match the static site domain, state-changing POST/PATCH requests were rejected with HTTP 403.
- **Implemented Resolution:** Updated cookie options in backend/src/routes/auth.js: sameSite: process.env.NODE_ENV === "production" ? "none" : "lax" with secure: true. This allows modern browsers to safely attach the session cookie across domains.

3. CORE ENHANCEMENTS IMPLEMENTED (ALIGNED TO PROPOSAL)

- **Admin Operational & Revenue Reports:** Upgraded backend/src/routes/administrator.js to compute live Total Revenue (month-to-date) and Average Turnaround Time (TAT from payment to release). Integrated dynamic cards and tables into Reports.jsx and AdminDashboard.jsx, complete with Excel/CSV export and print stylesheets.
- **Patient Portal PDF Result Export:** Implemented an official Diagnostic Laboratory Report view in MyResults.jsx featuring laboratory letterhead, test parameters, reference ranges, abnormal flags, pathologist remarks, and electronic authorization. Added print styles (@media print) for direct PDF generation and printing.
- **Multi-Analyte Diagnostic Panel Builder:** Transformed the administrator test catalog form (TestCatalog.jsx) to support multi-parameter panels (e.g. Complete Blood Count indices, Urinalysis) with custom units, options, and validation. Added a "Params" inspector modal backed by a new endpoint: GET /api/admin/tests/:id/configuration.
- **Event-Driven Cashier Queue Calling:** Enhanced CashierQueue.jsx to assign sequential Queue Numbers (Q-001, Q-002, ...), introduced a "Call Next Patient" action, and integrated a prominent "Now Calling" status banner to satisfy the proposal queue-calling requirement.
- **Zero-Fee & Waiver Request Processing:** Modified payments check constraint in backend/supabase/schema.sql from "amount > 0" to "amount >= 0". Cashier queue now provides a "Confirm Request" path for complimentary or fee-waived screenings.

4. DIAGNOSTIC TEST CATALOG STANDARD (12 CATEGORIES, 77 TESTS)

The repository includes standard-test-result-design.md, specifying 12 standardized clinical categories and 77 unique diagnostic tests to guide catalog initialization and reference interval modeling:

- **Hematology (8 tests):** Complete Blood Count (CBC), Differential WBC, Platelet Count, Hgb, Hct, ESR, Reticulocytes
- **Clinical Chemistry (12 tests):** FBS, RBS, HbA1c, BUN, Creatinine, Uric Acid, Total Protein, Albumin, Electrolytes (Na/K/Cl/Ca)
- **Lipid Profile (4 tests):** Total Cholesterol, HDL Cholesterol, LDL Cholesterol, Triglycerides
- **Liver Function Tests (7 tests):** ALT/SGPT, AST/SGOT, Alkaline Phosphatase (ALP), Total Bilirubin, Direct Bilirubin
- **Renal Function (6 tests):** eGFR, Urine Albumin-to-Creatinine Ratio (UACR), Urine Total Protein, Creatinine Clearance
- **Clinical Microscopy & Parasitology (7 tests):** Routine Urinalysis, Routine Fecalysis, FOBT, Pregnancy Test, Semen Analysis, Parasite Exam
- **Microbiology (7 tests):** Gram Stain, Acid-Fast Bacilli (AFB), KOH Mount, Bacterial Culture, Culture & Sensitivity
- **Immunology & Serology (7 tests):** CRP, Rheumatoid Factor, Dengue NS1 & IgM/IgG, HBsAg, HIV Screening, Syphilis
- **Coagulation (5 tests):** Prothrombin Time (PT), INR, aPTT, Fibrinogen, D-dimer
- **Hormones & Endocrinology (8 tests):** TSH, FT4, FT3, Total T4/T3, Cortisol, Testosterone, Prolactin
- **Blood Banking & Transfusion (5 tests):** ABO Grouping, Rh(D) Typing, Crossmatching, Direct/Indirect Antiglobulin Tests
- **Histopathology & Cytology (4 tests):** Pap Smear, Tissue Biopsy Examination, Cytology Examination, FNAC

  Implementation Rule: Canonical analytes (Albumin, Total Protein, Creatinine, BUN) prevent redundant billing codes. Reference intervals remain empty until calibrated to local clinic analyzer methods.

5. FILE MODIFICATION & VERIFICATION MATRIX

- **backend/src/routes/auth.js:** Updated cookie options to SameSite=None and Secure=true in production.
- **backend/src/routes/administrator.js:** Added Revenue & TAT metrics to summary; added test configuration query endpoint.
- **backend/supabase/schema.sql:** Relaxed payments constraint to allow amount >= 0 for fee-waived requests.
- **frontend/src/api/client.js:** Registered testConfiguration API helper method.
- **frontend/src/pages/administrator/AdminDashboard.jsx:** Added Revenue & TAT summary cards; wired dynamic turnaround display.
- **frontend/src/pages/administrator/Reports.jsx:** Surfaced Revenue & TAT indicators in summary, performance table, and Excel export.
- **frontend/src/pages/administrator/TestCatalog.jsx:** Built dynamic multi-analyte parameter builder and parameter inspection modal.
- **frontend/src/pages/administrator/TestCatalog.css:** Added styling for parameter cards, grids, and configuration views.
- **frontend/src/pages/cashier/CashierQueue.jsx:** Integrated queue numbers, "Call Next" action, and "Now Calling" status banner.
- **frontend/src/pages/cashier/CashierQueue.css:** Styled gradient calling banner, queue badges, and status pills.
- **frontend/src/pages/patient/MyResults.jsx:** Added Diagnostic Laboratory Report view with printable layout and release notice.
- **frontend/src/pages/patient/MyResults.css:** Configured @media print rules to isolate diagnostic report for clean PDF export.

6. QUALITY ASSURANCE & VERIFICATION RESULTS

- **Static Code Analysis (oxlint):** Evaluated across 74 frontend modules: 0 errors.
- **Production Bundler (vite build):** Transformed 139 modules and generated optimized production assets in 921ms without warnings.
- **Database Integrity:** Triggers, assignment validations, and clinic-scoped tenant isolation confirmed intact.